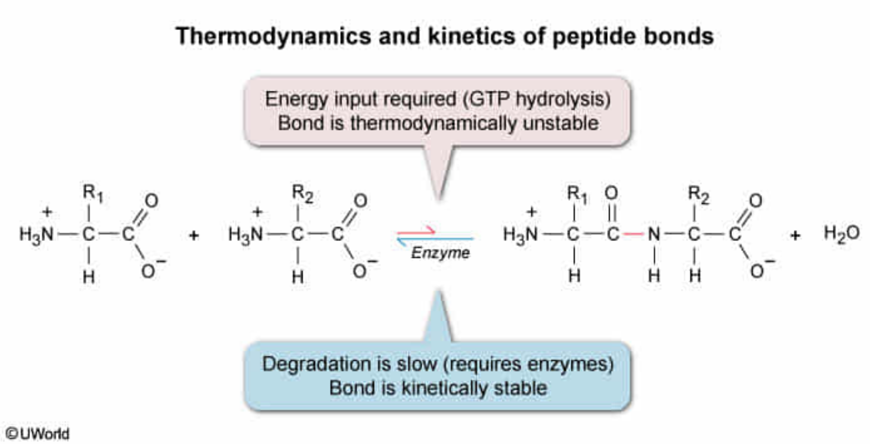
Under physiological conditions, peptide bond formation requires coupling to GTP hydrolysis. Peptide bond degradation does not require GTP hydrolysis but does require a protease enzyme to proceed at physiologically relevant rates. Based on this information, under physiological conditions and in the absence of proteases, the peptide bonds that form during protein synthesis are:

- A. both thermodynamically and kinetically stable. [19%]
- B. thermodynamically unstable but kinetically stable. [37%]
- C. thermodynamically stable but kinetically unstable. [34%]
- D. both thermodynamically and kinetically unstable. [8%]

Correct
37% answered correctly

Explanation:



The thermodynamic stability of a bond refers to its tendency to form or break in chemical reactions. A thermodynamically stable bond is a bond that forms during a spontaneous reaction (ie, ΔG is negative, no energy input is needed). In contrast, a thermodynamically *unstable* bond forms during nonspontaneous reactions. These reactions require energy input from another source to proceed (ie, positive ΔG), and the resulting bonds can be broken spontaneously by the reverse reaction.

In biological systems, the energy used to form thermodynamically unstable bonds often comes in the form of ATP or GTP hydrolysis. The question states that peptide bond *formation* requires coupling to GTP hydrolysis (energy input), suggesting that the **resulting peptide bond** is **thermodynamically unstable** and can therefore *break* in a spontaneous reaction.

However, spontaneous reactions do not necessarily proceed quickly. The rate of a reaction is described by its activation energy; a lower activation energy yields a faster reaction rate. Bonds that break in **slow reactions** (ie, reactions with *high* activation energies) are said to be **kinetically stable**. In biological systems, kinetic stability can allow thermodynamically unstable molecules to remain intact long enough to perform their biological functions.

The question indicates that peptide bond degradation occurs at a physiologically relevant (ie, fast) rate *only if* a protease enzyme is present. Protease enzymes catalyze the degradation reaction by decreasing its activation energy. However, in the *absence* of proteases, peptide bond **degradation is slow**, indicating that the **peptide bond is kinetically stable**. This stability allows proteins under physiological conditions to remain intact for significant periods of time despite their thermodynamic instability.

(Choices A, C, and D) Because peptide bond formation requires coupling to an energy source (ie, GTP hydrolysis), peptide bonds are thermodynamically unstable. Because peptide bonds do not break at a significant rate unless an enzyme is present, they are kinetically stable.

Educational objective:
Thermodynamically unstable bonds break spontaneously but not necessarily quickly. Bonds that break slowly in the absence of an enzyme are kinetically stable. Kinetic stability allows thermodynamically unstable molecules such as proteins to remain intact for long periods of time.

Time spent:0 sec
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Subject:
Biochemistry
System:
Amino Acids and Proteins